

WAVE OPTICS

Newton's corpuscular Model theory

- Very small particle emitted by light source (corpuscular particle)
- That particle moves with speed of light in straight line.
- Denser medium, speed of light more (This is totally wrong)
- Different colour due to diff. size of corpuscular particles.

* Explain:- Rectilinear propagation, Reflection, Refraction

* Fail to explain:- Interference, diffraction and polarization.

Huygens wave theory

- Assume light is a Mechanical & longitudinal wave.
- Requires medium for propagation.
- It gives concept of wavelet & wavefront.

* Explain:- Rectilinear propagation, Interference (YDSE), Reflection, Refraction, Diffraction.

* Fail to explain:- Polarisation, Photoelectric effect, Compton effect.

* Wave Equation → (A/q to Hugen)

$$y = A \sin(kx + \omega t + \phi) \rightarrow \text{light / sound / EM wave / AC same}$$

~~See above~~

$$\frac{2\pi}{\Delta\phi} = \frac{\lambda}{\Delta x} = \frac{T}{\Delta t} \rightarrow \Delta\phi = \left(\frac{2\pi}{\lambda}\right) \Delta x$$

Light source

① Frequency

- No. of oscillation per sec
- Source dependent, medium independent
- Unit → Hz = s⁻¹

$$T = \frac{1}{f}$$

② Angular frequency

$$\omega = 2\pi f \text{ rad/sec}$$

$$\omega = \frac{2\pi}{T}$$

③ Amplitude

- Max displacement from mean position on either side.
- depends on source & medium.

④ wavelength → The distance b/w the two successive crests or trough.

⑤ speed of wave (medium dependent)

$$v = \lambda f = \frac{\lambda}{T} = \frac{\omega}{k} = \frac{c}{\mu} = \frac{1}{\sqrt{\epsilon\mu}} = \frac{c}{\sqrt{\epsilon_r \mu_r}}$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$v = \frac{1}{\sqrt{\epsilon_r \mu_r}} = \frac{c}{\sqrt{\epsilon_r \mu_r}}$$

Wave front

- The locus of all points (wavelet) which oscillate in a same phase.
- The point of disturbance (oscillation) due to propagation.
- A line ⊥ to a wavefront is a ray.
- wave travel ⊥ to wavefront.
- Electric field (wave ray)
- Equipotential surface (wavefront)

Behaviour of plane wavefront on

Reflection & Refraction:- (do before your self)

- ① plane wave reflected from plane mirror.
- ② Refraction of a plane wavefront by a convex lens.
- ③ Refraction of a plane wavefront by concave lens.
- ④ Refraction of plane wave from prism.
- ⑤ Refraction of plane wave by concave mirror.
- ⑥ Planar shape of the wavefront on earth due to sun.

Point source	Linear source	Planar source
Spherical wavefront	Cylindrical wavefront	Planar wavefront
$I \propto A^2 \propto \frac{1}{r^2}$	$I \propto A^2 \propto \frac{1}{r}$	$I \propto A^2 \propto r^0$
$A \propto \frac{1}{r}$	$A \propto \frac{1}{\sqrt{r}}$	$A \propto r^0$

Intensity of wave

$$I = \frac{1}{2} \rho v A^2 \omega^2$$

ρ = density of Medium

v = velocity of wave

A = Amplitude

ω = Angular Velocity

*4r point

- The angle between incident wavefront & surface is angle of incident.
- The angle b/w surface & reflected wavefront is angle of reflection (r).
- Time b/w two same type of wavefront is always same.

Secondary wavelets (source)/Huygens construction

- ① Each & every point of wavefront behave as a secondary source.
- ② Those point (wavelet) creates spherical wavefront which transverse in forward direction with speed.
- ③ The secondary wavelets speeds in all directn but in backward directn intensity is zero.

Huygen construction

↳ A common tangent enveloping secondary wavelet in forward direction give new wavefront.

Demerits of Huygens wave theory

- Unable to explain polarization of light which can be only explain by transverse nature.
- He can't explain electromagnetic nature of light.
- why secondary wavelets are formed in forward direction.

Interaction of light

Ray $\rightarrow$ object $\gg \lambda$ light

wave $\rightarrow$ object $\approx \lambda$ light

Particle $\rightarrow e^-$ / proton / α -particle (photon)

Source

$\Rightarrow$ Phase difference

Time of $\frac{1}{f}$ constant coherent frequency = same	Time of $\frac{1}{f}$ variable Incoherent
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Coherent sources

$$\Delta\phi = 2\pi(\omega_1 - \omega_2)t + \Delta\phi'$$

If $\omega_1 = \omega_2$, $\Delta\phi = \text{constant}$; same λ
same v (frequency)

Incoherent sources

$$\Delta\phi = [\omega_1 - \omega_2]t + [k_1x_1 - k_2x_2] + [\phi_1 - \phi_2]$$

Time dependent term!

Huygen's principle

- Every point on the wave front becomes a source of secondary disturbance & generates wavelets which spread out in the medium with the same velocity as that of light in forward direction only.
- The envelope of these secondary waves at any instant of time gives the position of the new wave front at that instant.
- The wave front in medium is always perpendicular to the directn of wave propagation.

Interference

↳ Superposition of coherent waves
for incoherent source, $I = I_1 + I_2$
for coherent waves

$$I_{\max} > I_1 + I_2 \text{ (Bright)}$$

$$I_{\min} < I_1 + I_2 \text{ (Dark)}$$

$$y_1 = A_1 \sin(\omega t + \phi)$$

$$y_2 = A_2 \sin(\omega t + \phi_2)$$

Also, let $\Delta\phi$ be phase difference

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\Delta\phi)}$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\Delta\phi)$$

$$A_{\max} = A_1 + A_2, I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{\text{res}} = 4I_0 \cos^2(\Delta\phi/2)$$

→ If two intensities are same → Better contrast

Constructive

- $I_{\text{max}} = 4I_0$
- $A_{\text{max}} = A_1 + A_2$
- $I_{\text{max}} = (A_1 + A_2)^2 = (\sqrt{I_1} + \sqrt{I_2})^2$

$$\Delta\phi = 2\pi n \rightarrow \text{even}$$

$$\Delta x = n\lambda$$

Destructive

- $I_{\text{min}} = 0$
- $A_{\text{min}} = A_1 - A_2$
- $I_{\text{min}} = (A_1 - A_2)^2 = (\sqrt{I_1} - \sqrt{I_2})^2$

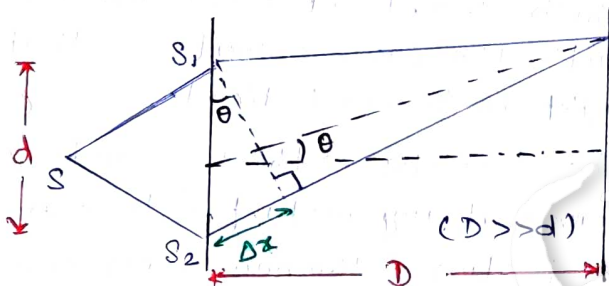
$$\Delta\phi = (2n-1)\pi \rightarrow \text{even}$$

$$\Delta x = (2n-1)\lambda/2$$

If intensities are different

$$\frac{I_{\text{max}}}{I_{\text{min}}} = \left[\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right]^2 = \left[\frac{A_1 + A_2}{A_1 - A_2} \right]^2$$

Young's double slit experiment



★ Two virtual sources (S_1 & S_2) [Actually pinholes] were used by young in the expt.

★ Strictly, the fringe pattern will be a Hyperbola.

$$\Delta x = d \sin \theta$$

$$\Delta\phi = 144\pi = \frac{2\pi}{\lambda} \Delta x$$

Angular position

$$\Delta x = d \left(\frac{y}{D} \right) = n\lambda$$

⇒ Distance of n th bright

$$y_b = n \left(\frac{\lambda D}{d} \right)$$

⇒ Distance of n th dark

$$y_d = \left(\frac{2n-1}{2} \right) \left(\frac{\lambda D}{d} \right)$$

⇒ Fringe width,

$$\beta = \frac{\lambda D}{d}$$

V DSE in other medium

$$\beta' = \frac{\beta}{\mu}$$

$$\lambda' = \frac{\lambda}{\mu}$$

frequency = constant

⇒ N-identical coherent sources

$$I_{\text{max}} = N^2 I_0$$

$$I_{\text{min}} = 0$$

⇒ On a given size of screen

$$n_1 \lambda_1 = n_2 \lambda_2$$

$$\text{or } n_1 \beta_1 = n_2 \beta_2$$

⇒ Angular fringe width

$$\theta = \frac{\beta}{D} = \frac{\lambda}{d}$$

⇒ fringe visibility

$$V = \frac{I_{\text{max}} - I_{\text{min}}}{I_{\text{max}} + I_{\text{min}}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$$

white light

- 1st maxⁿ sabse pehle violet (n) 3-4 cm
- Sabse pehle Red dikhega.
- Used to find central maxima.

VIBGYOR

→ $\lambda \uparrow$

$$\beta \propto \lambda$$

$\beta_{\text{red}} > \beta_{\text{yellow}}$

⇒ Position of central maxⁿ with glass of thickness 't'.

$$y = \frac{Dt(\mu-1)}{d} \therefore \lambda = (\mu-1)t$$

No. of bright & dark fringes

$$\text{Bright fringe} = \left[\frac{2d}{\lambda} + 1 \right]$$

$$\text{Dark fringe} = 2 \left[\frac{d}{\lambda} + \frac{1}{2} \right]$$

Condition for sustained substance

- Two source must be monochromatic & coherent
- Two source ~~must~~ should be coherent; they could vibrate in the same phase, must have constant phase difference.

- Both must have same frequency.
- The separation b/w coherent source must be small.

Diffraction

* Bending of light waves around the edges of small obstacles.

* Mechanical waves
EM waves
(All types of waves) } **Diffraction**

* larger $\lambda \Rightarrow$ Easier diffraction.

$$\text{width} = \frac{2\lambda D}{a}$$

$$\text{Angular size} = \theta = \frac{2\lambda}{a}$$

$$\Delta x = a \sin \theta = a \tan \theta = n\lambda$$

Minima (D.I)

$$a \sin \theta = n\lambda$$

$$n = 1, 2, 3, 4$$

$$\sin \theta = \tan \theta = \frac{y}{D}$$

Maxima (C.I)

$$a \sin \theta = (2n-1) \frac{\lambda}{2}$$

$$n = 2, 3, 4$$

* Types of Diffraction

① Fresnel

- * Source & screen at finite distⁿ.
- * Spherical wave front

② Fraunhofer

- * Source & screen at ∞ distance
- * planar wave front

Difference b/w interference & diffraction (for Fraunhofer single slit):-

⇒ Interference

- ① Superposition of 2 waves from 2 coherent sources.
- ② Bright lines are equally bright & equally spaced.
- ③ All dark lines are totally dark.
- ④ Bands are large in number.

⇒ Fraunhofer diffraction

- ① Superposition of 2 waves coming from two different parts of the same wave front.
- ② Brightness goes ↓ with ↑ in angle of diffraction.
- ③ dark lines are perfectly dark but contrast with bright goes ↓.
- ④ Diffraction bands are few in number

polarization

↳ Electromagnetic wave is E.M transverse wave, $\vec{E}$ & $\vec{B}$ are $\perp$ to each other & $\perp$ to wave.

↳ dirⁿ of E.M wave = $\vec{E} \times \vec{B}$

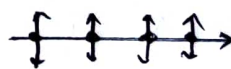
↳ $|\vec{E}| \gg |\vec{B}|$, so light is preferably described as the oscillation of electric field only.

Unpolarised light

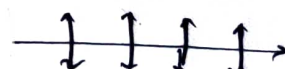
- An ordinary beam of light consists of a large no. of waves emitted by the atoms of the light source.
- Each atom produces a wave with its own orientation of $\vec{E}$, so all dirⁿ of vibration of $\vec{E}$ are equally probable.
- ordinary light vibrating in all dirⁿ in plane perpendicular to propagation.

Polarized light

- Light wave vibrating in a particular dirⁿ in plane $\perp$ to the propagation of light.



unpolarised



completely polarised



complete polarized

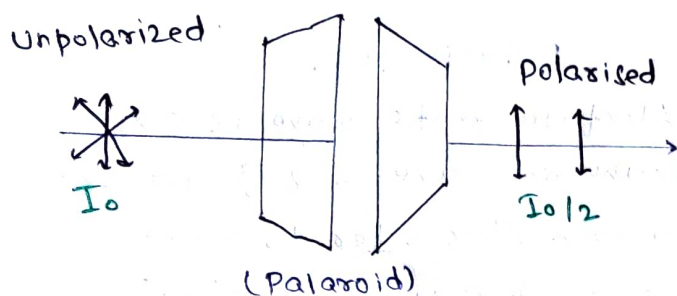


Partially polarized.

* When unpolarized light is passed through **Polariser** (Tourmaline, Nicol prism, Polaroid).

(i) The electric vector oscillating $\perp$ to aligned molecules ($\parallel$ to pass axis).

(ii) The electric vector in the dirn of aligned molecules get adsorbed.



* Plane of Vibration (POV)

→ The electric vector which passes through polaroid vibrates in a plane.

* Plane of Polarization (POP)

→ Plane $\perp$ to POV.

MALUS' LAW

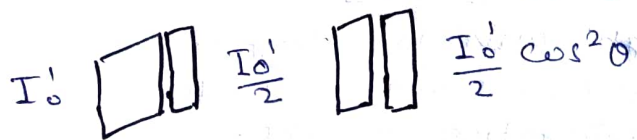
→ Two polaroids P_1 & P_2 with angle θ b/w their pass axes.

→ Let I_0 be the intensity of polarized light after passing through P_1 .

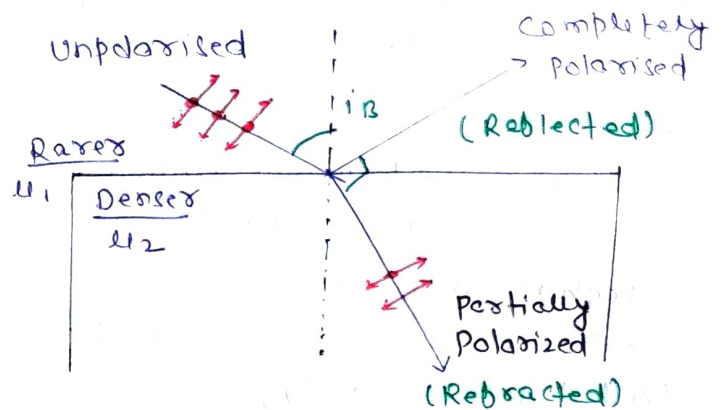
* Intensity of light through P_2

$$I = I_0 \cos^2 \theta$$

[P_1 = polariser P_2 - Analyser]



Polarisation by Reflection



* The reflected light is polarised with its electric vector $\perp$ to the plane of incidence when the refracted & reflected light are at right angles.

$$\frac{\mu_2}{\mu_1} = \mu_{21} = \tan i_B$$

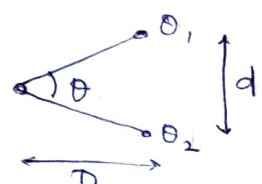
i_B → Brewster's angle

This is called Brewster law.

* If $i \neq i_B$, both reflected & refracted light are partially polarised.

* Resolving Power

Resolving limit
(or) limit of Resolution



$$\theta = \frac{d}{D}$$

$$\text{Resolving Power} = \frac{1}{\text{Limit of Resolution}}$$

Telegram → @CYANNOTES